Statistical Case Studies - Project 1 (Part 1)

Anna Scollay, Lewis Seath, Zacharias Onyejiaka

2026-10-12

Executive Summary

Data and Exploratory Analysis

Data Description and Quality

Table 1: Data used in this analysis, and the quality codes assigned to each reading

Data description	
Station	Don at Parkhill
Source	SEPA time series service (KiWIS)
Measurement	River discharge (m^3/s)
Resolution	15 minutes
Period	01 October 2016 – 30 September 2026
Readings	350,592
Missing readings	0
Quality codes	
50 – Good (G)	335,515 readings (95.70%), used
100 – Estimated (E)	14,915 readings (4.25%), used
140 – Provisional (Prov)	0 readings (0.00%), not used
150 – Suspect (S)	0 readings (0.00%), not used
200 – Unchecked (imported from legacy database) (V)	0 readings (0.00%), not used
254 – Unchecked (U)	162 readings (0.05%), not used

We chose the Don at Parkhill station, using 15-minute discharge data covering the ten water years [add link to definition maybe?] from 1st October 2016 to 30th September 2026. The record contains 350,592 readings, with none missing, and each reading carries a quality code assigned by SEPA, which shows that almost all of the data is classified as either Good or Estimated. We retain the Estimated readings since [recognised by SEPA? and are they concentrated around periods of high flow or anything?]. The 162 unchecked readings, which make up only 0.05% of the record, are removed because [accuracy not verified? and need to confirm that none of the ones we're removing fall near a flood peak or anything].

Exploratory Analysis

Before selecting a threshold, we explore the full dataset to understand how flow on the Don behaves, focusing on the rarity and seasonality of high flows, and on how individual flood events differ from one other.

Hydrograph and Distribution

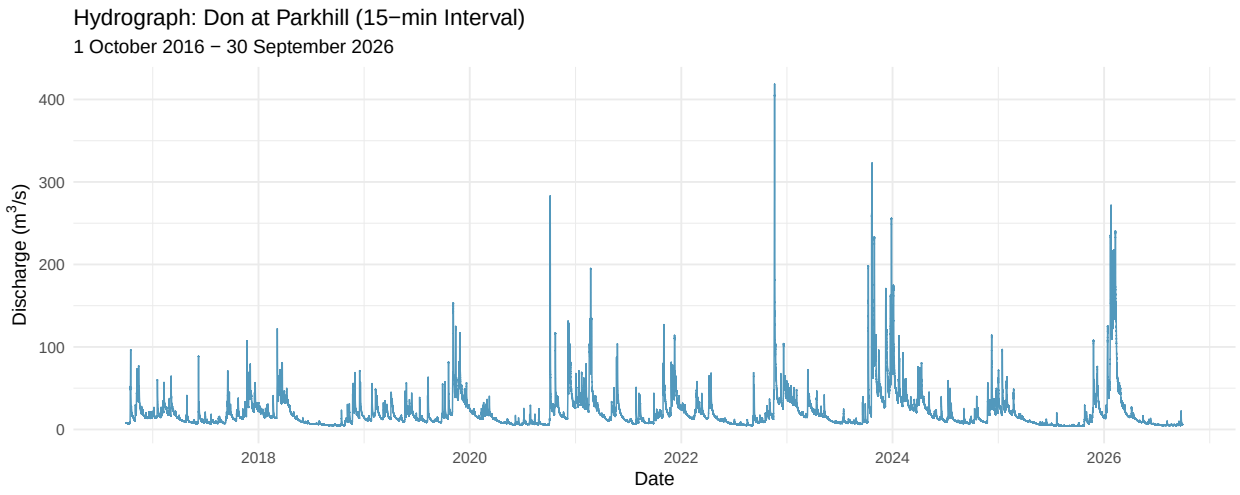
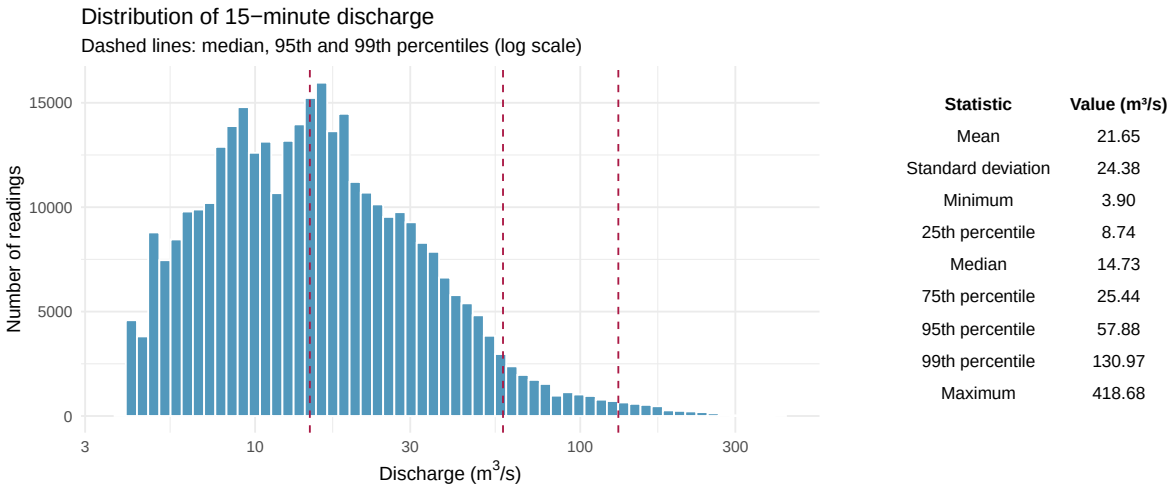


Figure 1: Hydrograph of 15-minute discharge for the Don at Parkhill.



Statistic	Value (m³/s)
Mean	21.65
Standard deviation	24.38
Minimum	3.90
25th percentile	8.74
Median	14.73
75th percentile	25.44
95th percentile	57.88
99th percentile	130.97
Maximum	418.68

Figure 2: Distribution of 15-minute discharge on a log scale, with summary statistics. Dashed lines mark the median, 95th and 99th percentiles.

From Figure 1, we observe that the discharge is low for most of the year, with a smaller number of large peaks, the largest reaching 418.68 m^3/s on 19th November 2022. The distribution in Figure 2 is strongly right-skewed, with a median of 14.73 m^3/s compared to a 99th percentile of 130.97 m^3/s, which initialises our view that extreme flows behave very differently from typical flows and need to be analysed seperately.

Seasonality of High Flows

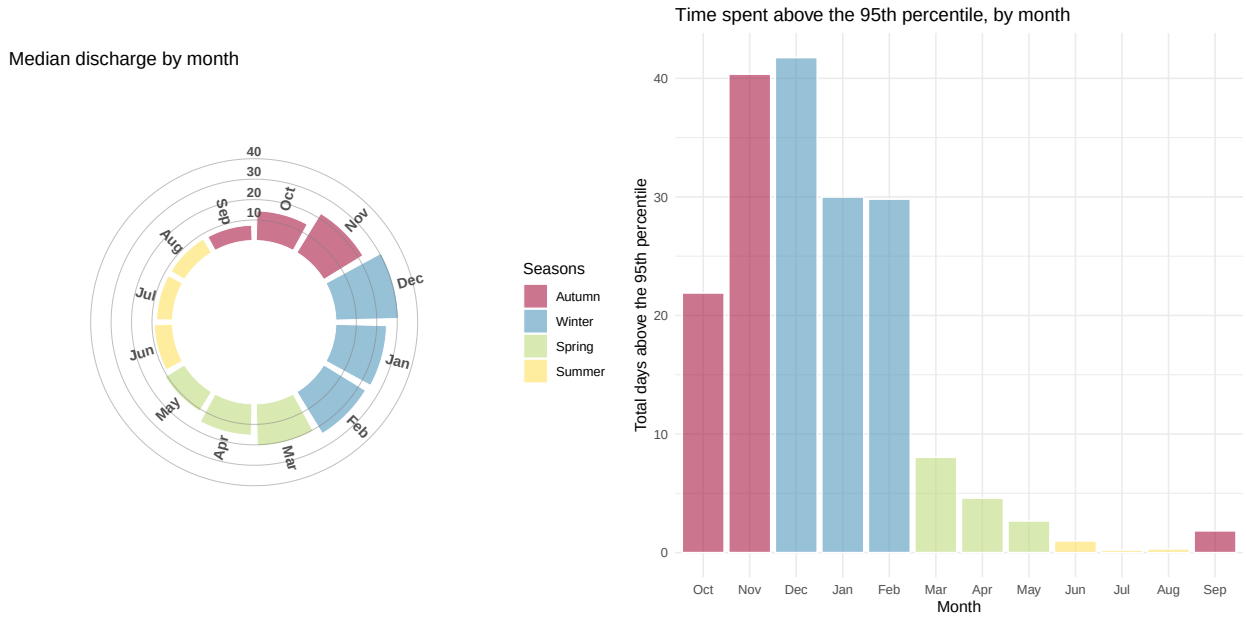


Figure 3: Median discharge by month (left) and total days spent above the 95th percentile in each month (right).

We observe that median discharge is highest from December to February and lowest from July to September, while time spent above the 95th percentile is even more concentrated to between October and February.

Flood Event Hydrographs

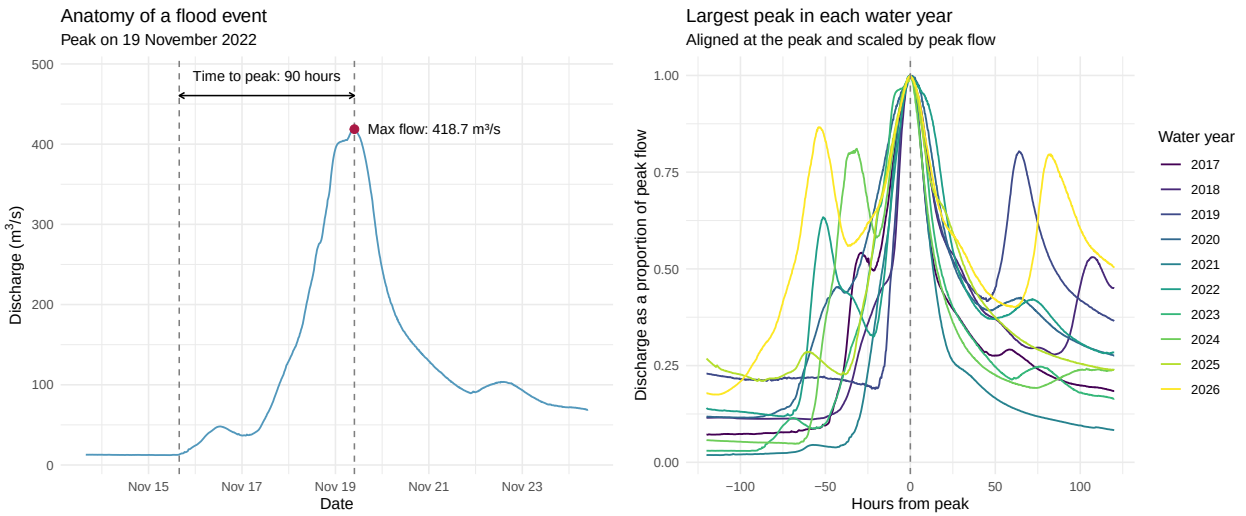


Figure 4: The largest recorded flood event, showing the max flow and time to peak (left), and the largest peak in each water year, aligned at the peak and scaled by peak flow (right).

This introduces the two quantities we are investigating, max flow and time to peak, and a key assumption we use throughout the report:

- Start of the rise: the last reading before the peak with flow at most $q_{\min} + 0.05\,(q_{\text{peak}} - q_{\min})$, where $q_{\min}$ is the lowest flow in the previous 7 days.

The left panel of Figure 4 shows the two quantities for the largest observed event, and in the right panel we observe that the largest peak in each water year varies considerably in shape, which suggests that the current approaches to flood inundation modelling of treating the hydrograph shape as fixed ignores an important source of uncertainty. Additionally, we observe the number of peaks clustered around each largest peak, highlighting the importance of considering independent events.

Summary

Since high flows are rare, concentrated between October and February, and vary in shape between events, we need a threshold that extracts them. We will then separate peaks into independent events, before investigating how max flow and time to peak vary across events and between seasons.

Threshold Selection and Independence

Initial Threshold

The Peaks Over Threshold (POT) approach extracts flood events as the peaks of flow above a threshold u. Two common rules for choosing u are the 95th percentile of flow and a threshold giving 3 to 5 peaks per year. Figure X shows both on the hydrograph, where the 95th percentile is 57.9 m^3/s, and counting each period above the threshold as a peak gives at least 5, 4 and 3 peaks per year at 79, 93 and 113 m^3/s. At this stage, however, a single flood can cross the threshold several times, so these counts overstate the number of distinct events. Robson and Reed (1999) note that a threshold giving around five peaks a year leaves the flexibility to raise it later to exclude smaller floods, and the lecture notes that the threshold may need to change once independence has been applied. Since the 95th percentile gives 9.4, well above this range, we take the threshold giving 5 peaks per year $u = 79\ m^3/s$, as our initial threshold.

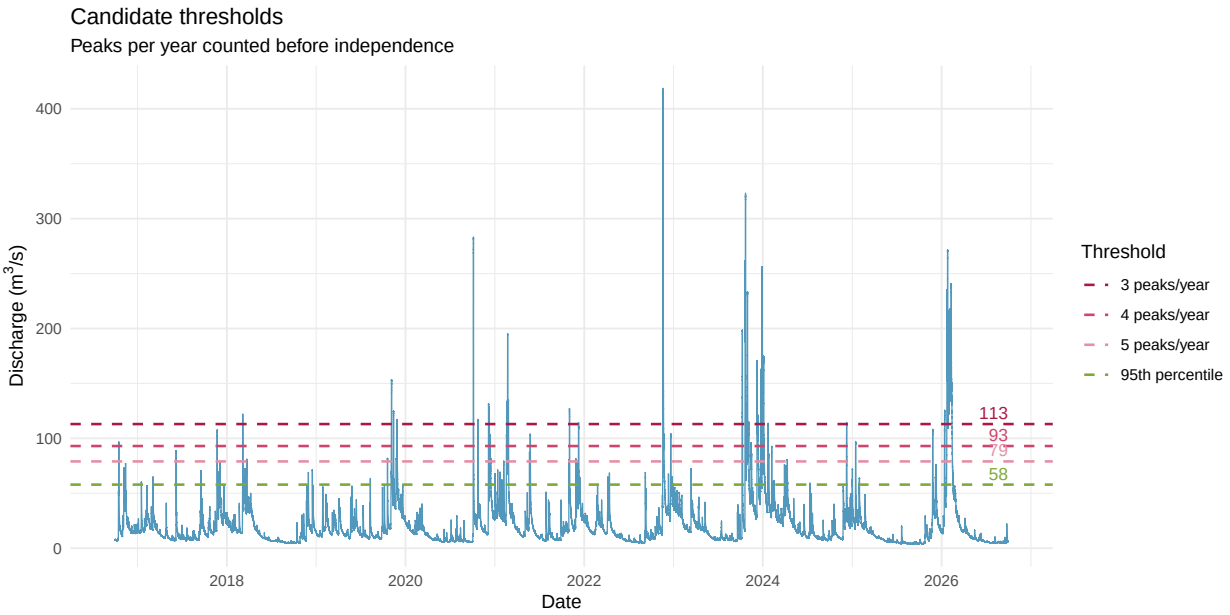


Figure 5: Hydrograph with the candidate thresholds. Peaks per year are counted before applying the independence rules.

Independence of Peaks

To make sure each peak represents a separate flood, we apply the independence rules of Robson and Reed (1999). Two adjacent peaks are only treated as separate events if:

1. they are at least three times the average time to rise apart, where the time to rise is the time between a trough and its peak, averaged over at least five clean peaks above the threshold; and
2. the flow at the trough between them is at least two-thirds lower than the first peak, i.e. below a third of it.

If either rule fails, we treat the peaks as one event and keep the larger peak. In applying these rules, we use the following assumptions:

- Clean peak: a peak with no other candidate peak in the previous 7 days.
- Trough: the start of the rise defined in the exploratory analysis, which we use to calculate the time to rise.

At the initial threshold, the average time to rise is 57.2 hours, so peaks must be at least 171.6 hours apart. This reduces the 56 candidate peaks to 35 independent events, or 3.5 per year (Figure X).

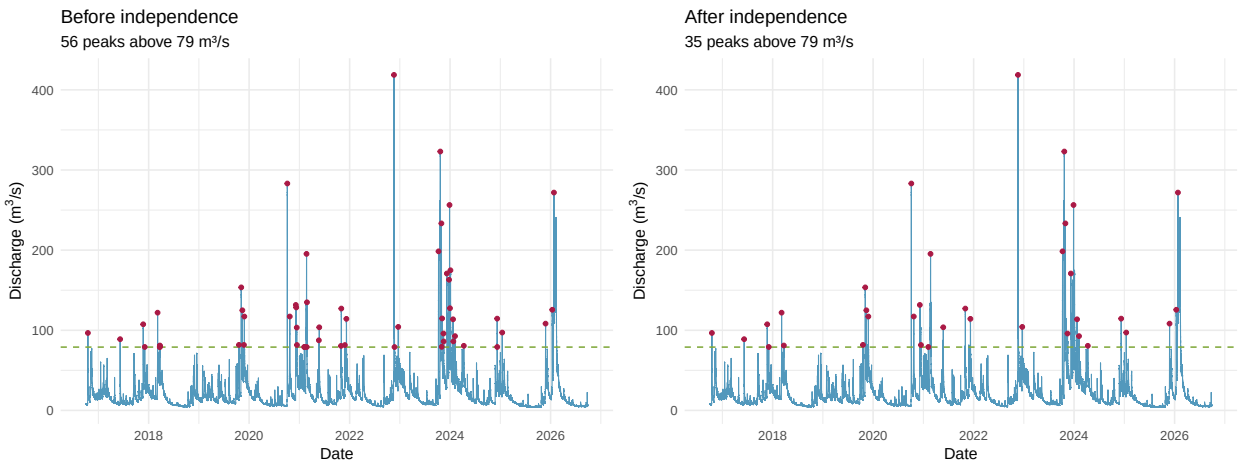


Figure 6: Peaks above the initial threshold before (left) and after (right) applying the independence rules.

Final Threshold

Since separating events lowers the number of peaks per year, we recheck the threshold using independent peaks only. Our initial threshold gives 3.5 independent peaks per year, while Figure X shows that thresholds between 57 and 81 m³/s give 3 to 5.

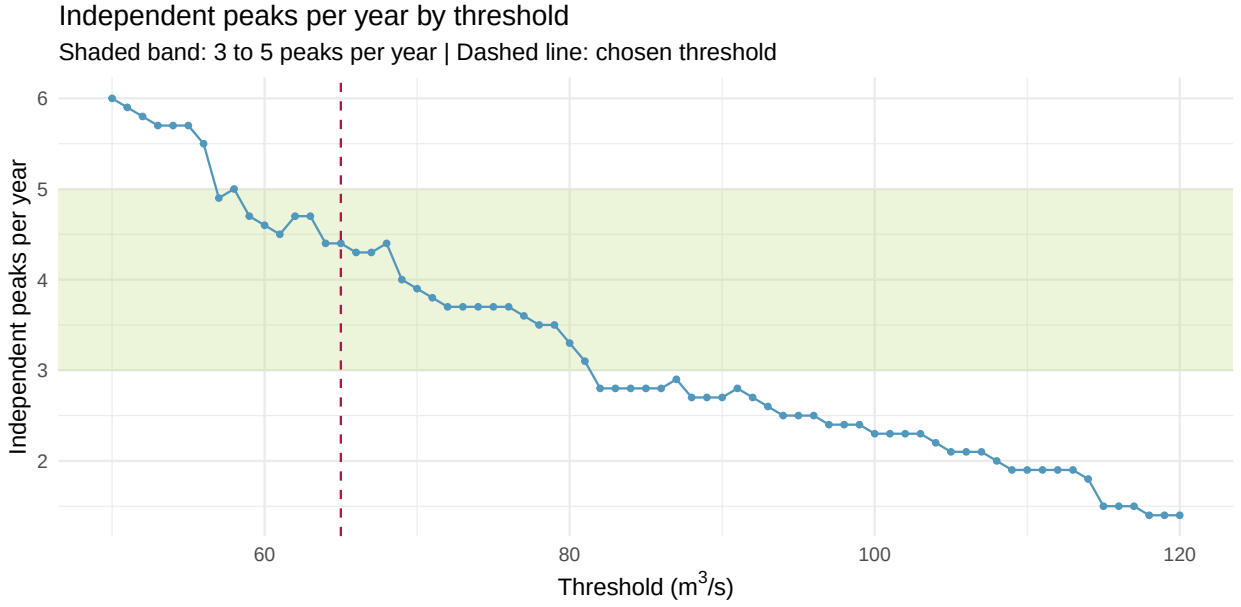


Figure 7: Independent peaks per year at each threshold.

To check which of these is suitable for extreme value modelling, we fit a Generalised Pareto Distribution (GPD) to the excesses of the independent peak flows X above u,

$$P(X - u \le y \mid X > u) = 1 - \left(1 + \frac{\xi y}{\sigma}\right)^{-1/\xi}, \quad y > 0, \quad 1 + \frac{\xi y}{\sigma} > 0,$$

where $\sigma > 0$ is the scale and $\xi \neq 0$ the shape parameter, with $\xi = 0$ giving an exponential tail as a limiting case. If the GPD holds above u, the mean excess $E[X - u \mid X > u] = (\sigma_0 + \xi u)/(1 - \xi)$ is linear in u (for

$\xi < 1$), where σ_0 is the scale parameter at a threshold of 0, and the estimates of ξ and the modified scale $\sigma - \xi u$ stay roughly constant as the threshold increases (Coles, 2001). [From Figure X, the mean residual life plot is approximately linear above ... m^3/s, and the parameter estimates are stable from ... m^3/s.]

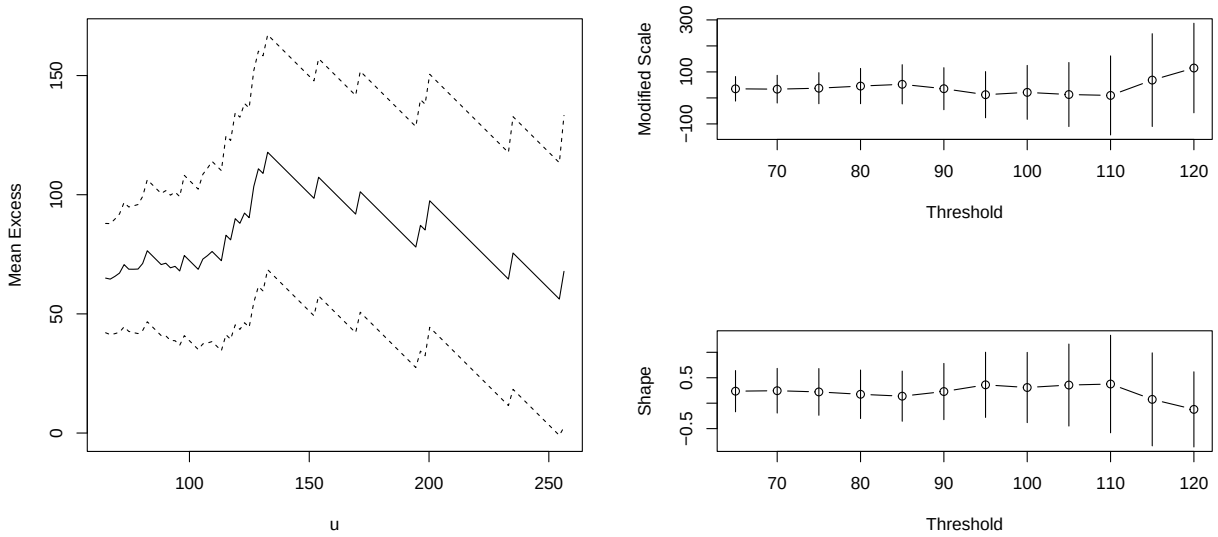


Figure 8: Mean residual life plot (left) and parameter stability plots (right) for the independent peaks at the final threshold.

Since the lecture notes that the threshold may need to change once independence has been applied, we lower it to a final threshold of $u = 65$ m^3/s, [which ...], giving 44 independent events (4.4 per year), with fitted parameters $\hat{\sigma} = 50.34$ (standard error 12.76) and $\hat{\xi} = 0.236$ (standard error 0.207). [The diagnostic plots in Figure X show ...]

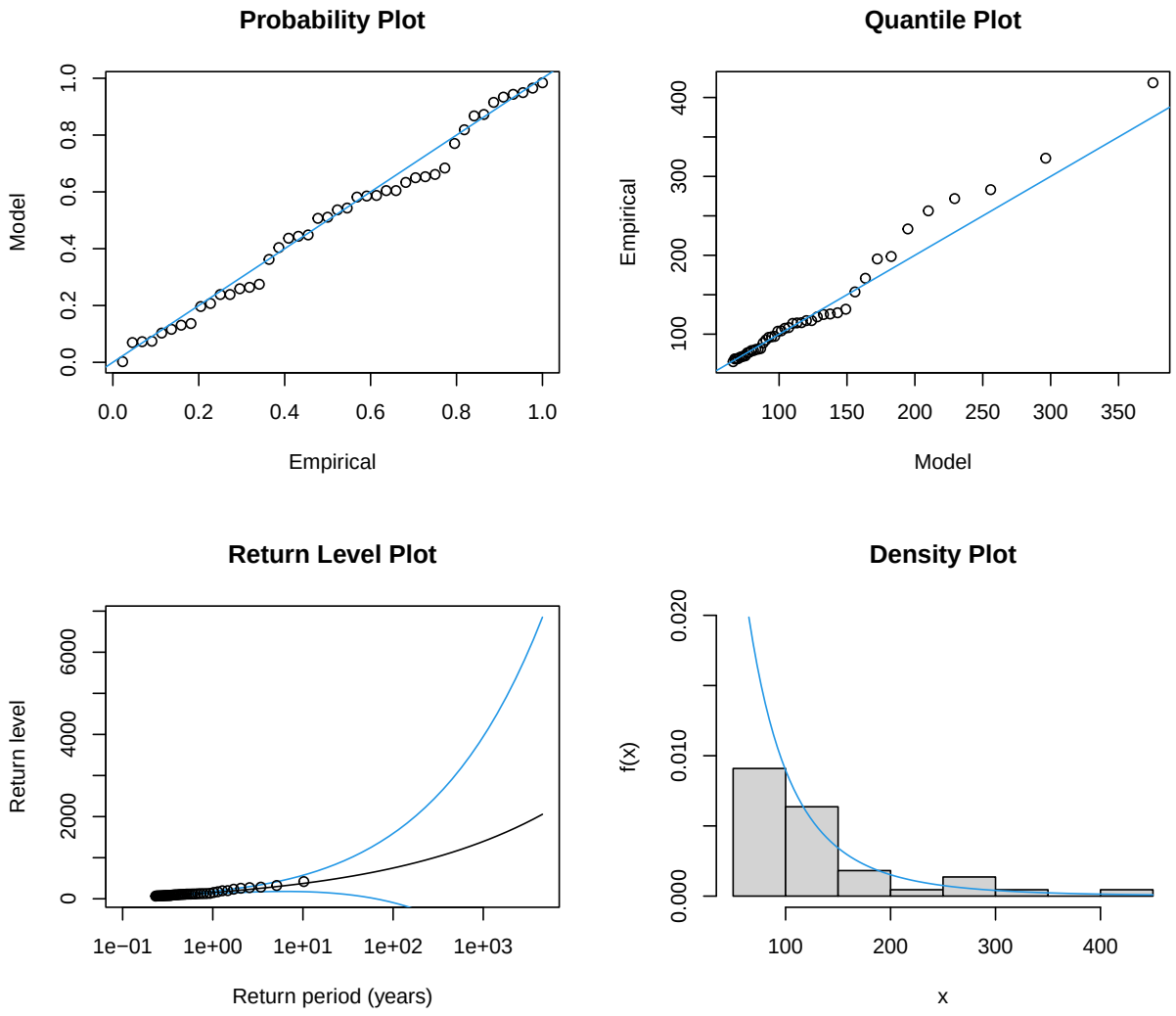
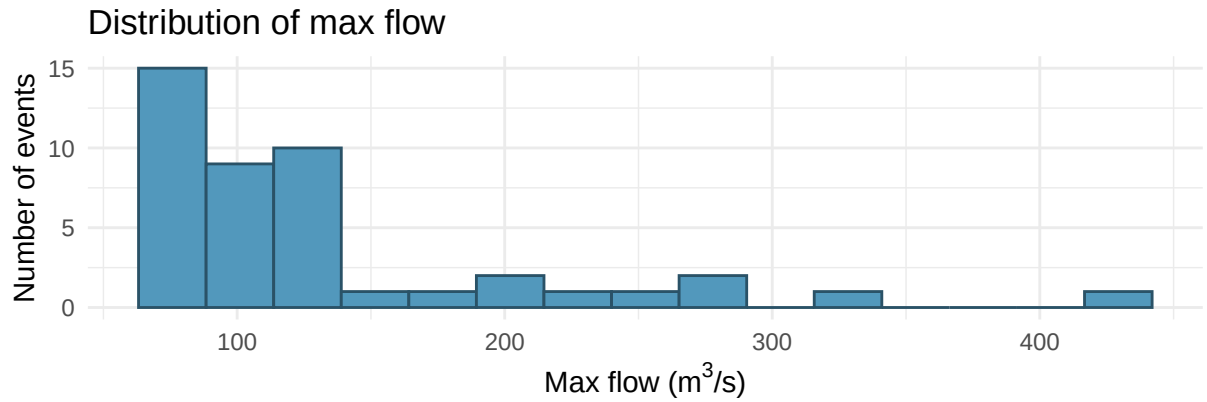
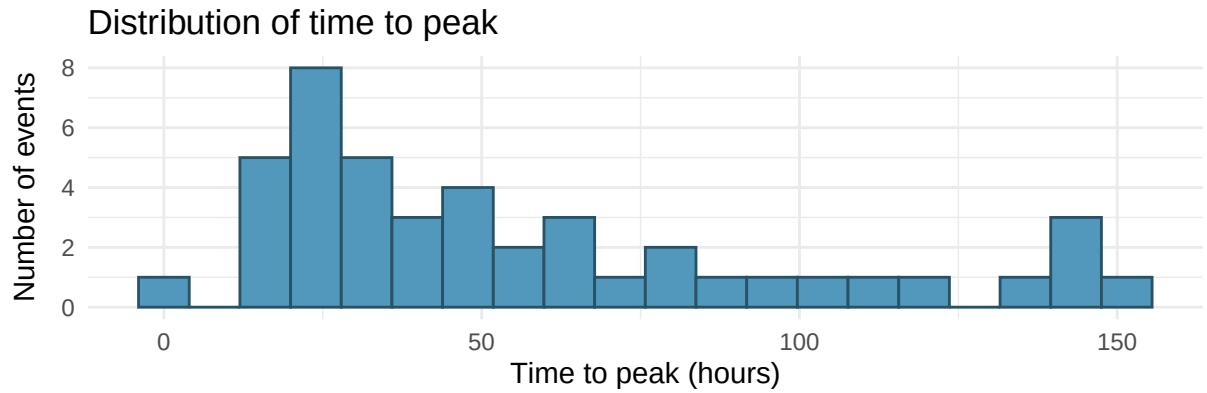
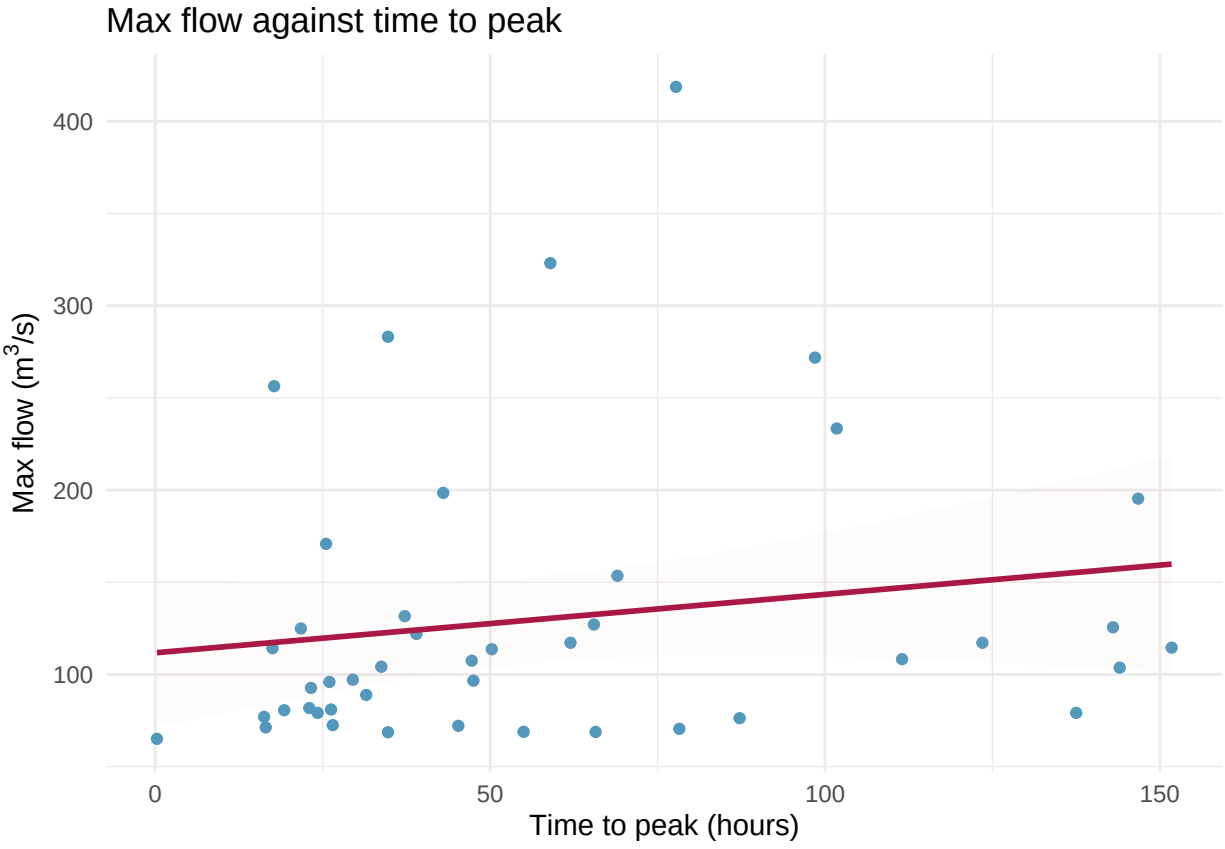


Figure 9: Diagnostic plots for the GPD fitted at the final threshold.

Uncertainty Analysis



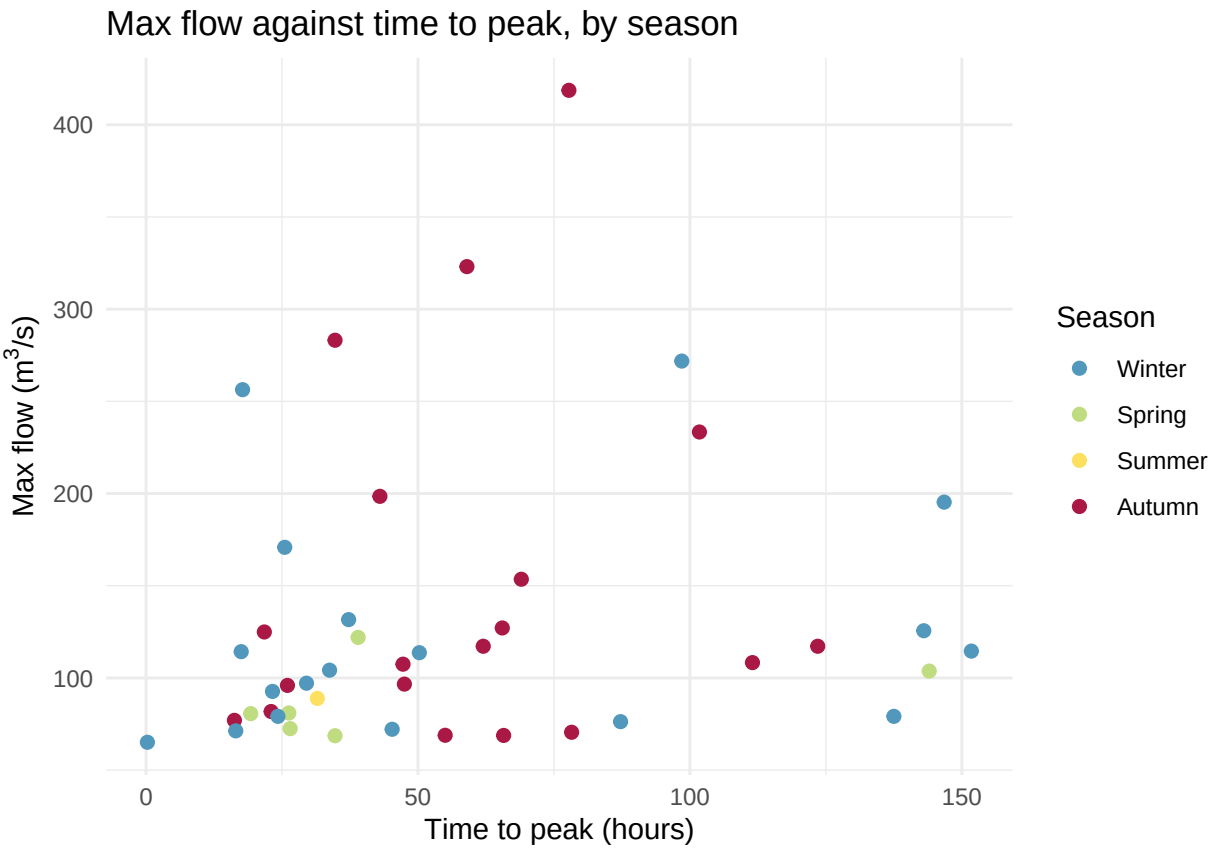


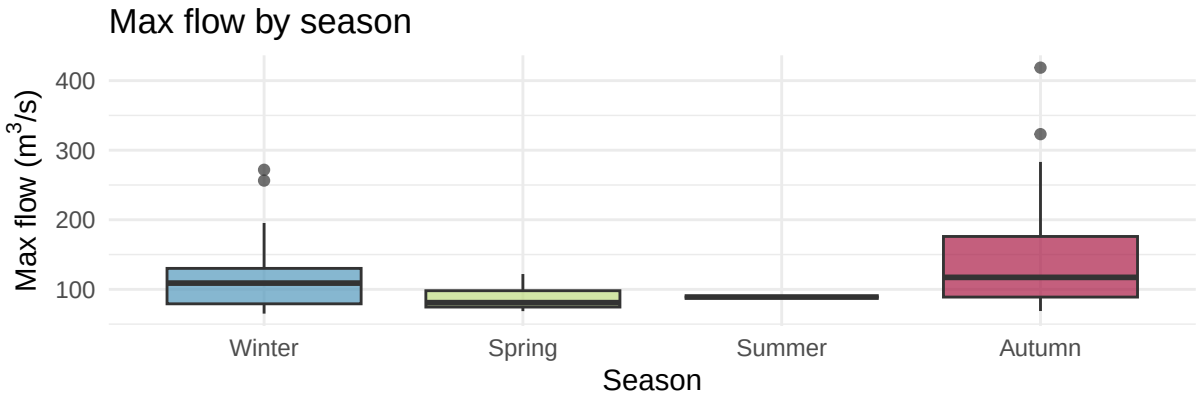


```
##
##  Spearman's rank correlation rho
##
## data:  peaks$time_to_peak_h and peaks$max_flow
## S = 10112, p-value = 0.05852
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.2874159
```

Table 2: Summary of Max Flow and Time to Peak for each Season

season	Events	Median max flow	Median time to peak (h)
Winter	18	108.951	35.500
Spring	6	80.812	30.625
Summer	1	88.889	31.500
Autumn	19	117.185	59.000





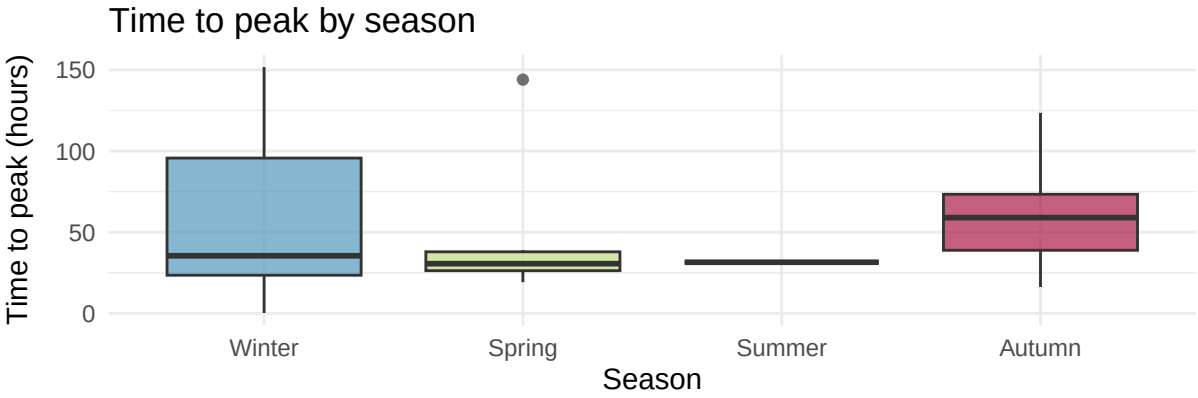


Table 3: Final Uncertanity Table

threshold	events	Events per Year	Median Max Flow	Median Time to Peak (Hours)	Spearman Rho Corelation
55	57	5.7	81.792	43.000	0.066
60	46	4.6	100.434	44.125	0.184
65	44	4.4	105.834	44.125	0.287
70	39	3.9	108.308	43.000	0.270
75	37	3.7	114.256	47.250	0.220